

1. (Spring 2024, #6) Find the sum of the series $\sum_{n=0}^{\infty} \frac{3(-1)^n}{2^{2n+1}}$.

2. (Spring 2020, #9) Consider the sequence (a_n) defined recursively by

$$\begin{cases} a_0 = 1 \\ a_{n+1} = \frac{-2}{2a_n + 5}, \quad \text{for } n \geq 0 \end{cases}$$

You may freely use the following **fact**: (a_n) is always contained in $\left(-\frac{5}{2}, \frac{7}{2}\right)$.

- (a) Compute the values of a_0 , a_1 , and a_2 .
- (b) Use mathematical induction to determine whether (a_n) is increasing or decreasing.
- (c) Show that (a_n) is convergent and determine the value of its limit. You must justify your findings.
- (d) Determine the convergence of the series $\sum_{n=0}^{+\infty} a_n$. Justify your answer.

3. (Fall 2019, #6a) Suppose a particle is moving back and forth along the x -axis. The particle begins at the point $x = 0$. It then moves in the positive x -direction a distance of 1. The particle then moves in the negative x -direction a distance of $1/4$. The particle then moves in the positive x -direction a distance of $1/16$. The particle then moves in the negative x -direction a distance of $1/64$, and so on. In general, whenever the particle moves a distance h in one direction, it then moves in the opposite direction a distance of $h/4$. What position on the x -axis does the particle eventually approach?

4. (Spring 2015, #6c,d)

(a) Find the limit if it exists of the sequence given by $u_n = \left(\frac{n+3}{n+1}\right)^n$.

(b) Is $\sum_0^{\infty} \left(\frac{n+3}{n+1}\right)^n$ convergent or divergent?

5. (Spring 2012, #8) Find the sum of the series $\sum_{n=2}^{\infty} \left[\frac{2}{n^2 - 1} + \left(\frac{2}{3} \right)^n \right]$.